SOLUTION ARCHITECTURE

Date	02 October 2026
Team ID	SWTID-2026-8237
Project Name	Comic Craft - AI Comic Story Creator using Gemini Models

Maximum Marks: 5 Marks

Solution Architecture

Comic Craft is a web-based AI comic generation application. The architecture connects the presentation layer with a Fast API backend and AI services. User-provided story details are processed to generate a structured comic outline, narration and dialogue, panel illustrations, a preview layout, and a downloadable PDF.

Solution Architecture Diagram

PRESENTATION / CLIENT LAYER
HTML + CSS + Jinja2 templates index.html → comic_preview.html → export_success.html

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API / APPLICATION ENTRY LAYER
Fast API routes.py /generate /generate-comic/json /test-image /export-success

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CORE LOGIC SERVICE
Comic generation workflow • generate _outline() • generate _story() • generate _image() • build _comic _layout() • save _pdf()

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EXTERNAL AI INTEGRATIONS
Google Gemini Flash → 5-panel outline Google Gemini Pro → narration & dialogue Stable Diffusion / Hugging Face Diffusers → illustrations

↓ ↑

DATA / STORAGE LAYER
static/panels → generated panel images static/exports → generated PDF files

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects story prompt, character name, setting, tone, and art style. Displays generated comic panels and export confirmation.	HTML, CSS, Jinja2
API Gateway / Application Layer	Receives form and JSON requests, routes comic-generation operations, handles errors, and connects frontend requests to the generation workflow.	FastAPI, Uvicorn, routes.py, Pydantic
Core Logic Service	Coordinates outline creation, story generation, image generation, panel layout construction, and PDF export.	Python, generate _outline(), generate _story(), generate _image(), build _comic _layout(), save _pdf()
External AI Integrations	Generates structured story content, detailed narration/dialogue, and comic-style illustrations from user requirements and generated prompts.	Google Gemini Flash, Google Gemini Pro, Stable Diffusion, Hugging Face Diffusers
Data / Storage Layer	Stores generated panel images and final exported PDF files for preview and download.	static/panels, static/exports, image files, PDF
PDF Export Component	Combines panel images and narrative content into a multi-page downloadable comic PDF.	FPDF

Architecture Flow

1. User enters the story prompt, character name, setting, tone, and art style in index.html.

2. Fast API receives the form request through the /generate route or a JSON request through /generate-comic/json.

3. Gemini Flash generates a structured 5-panel comic outline containing panel number, title, scene description, and image-generation prompt.

4. Gemini Pro expands the outline into comic narration and character dialogue.

5. Stable Diffusion generates an illustration for each panel using the generated image prompt.

6. build _comic _layout() associates each generated image with its corresponding panel story.

7. comic_preview.html displays the completed comic panel-by-panel.

8. save _pdf() uses FPDF to compile the comic into a multi-page PDF saved in static/exports.

9. export_success.html confirms successful export and allows the user to create another comic.

Architecture Summary

The Comic Craft architecture follows a layered flow from user interface to backend processing, AI integration, and file storage. The frontend is implemented with HTML, CSS, and Jinja2, while Fast API manages routing and application processing. Gemini Flash and Gemini Pro handle different stages of text generation, and Stable Diffusion handles image generation. The resulting panels are organized and exported as a PDF through the FPDF-based export component.

Source: Comic Craft project document and supplied Solution Architecture template